

Design of a Production Line Based on Lean Manufacturing and Simulation for the Ecuadorian Manufacturing Industry

- Mateo Sebastián Echeverria Peña
- Advisor: Ing. Juan Carlos Llivisaca Villazhañay, M.Sc.

Abstract

A manufacturing line was redesigned using Lean Manufacturing tools and discrete-event simulation (FlexSim), identifying its main constraint and validating the proposed improvements through a statistical t-test.

Objetives

General Objective

To evaluate the technical feasibility of implementing a specialized layout by integrating Lean Manufacturing and simulation under a demand growth scenario.

Specific Objectives

To analyze the current production system and relevant literature in order to identify technical, operational, and capacity requirements for a specialized production line.

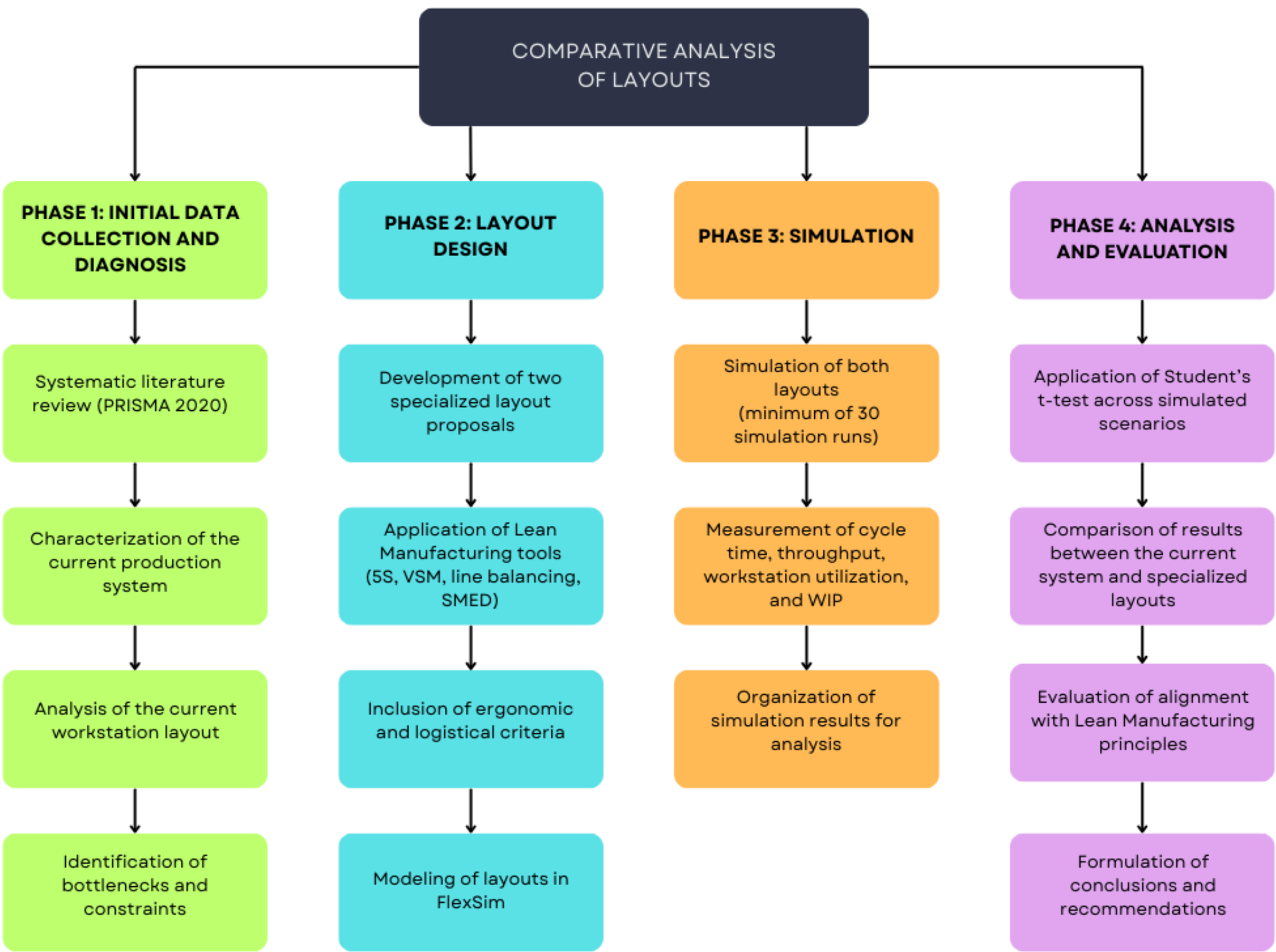
To design a specialized Lean Manufacturing principles, aimed at improving process flow.

To simulate the dynamic behavior of the proposed production line using FlexSim.

To evaluate the technical feasibility of the specialized line through a comparative analysis of simulation results against the current system.

Specific Objectives

Methodology



Curricular Integration

- Production Simulation
- Logistics and Supply Chain Management
- Lean Manufacturing (I and II)
- Experimental Design
- Research Methodology

Applied Engineering

Standards and Guidelines Applied

- PRISMA 2020: systematic literature review
- ISO 9001:2015: process-based approach and continuous improvement
- ISO 6385:2016: ergonomic criteria in design
- ISO 22400: manufacturing KPI framework

Design Constraints (Real Conditions Considered)

- No physical implementation, validation through simulation
- Aggregated and normalized operational data due to confidentiality

Keywords

Lean Manufacturing

Simulation

FlexSim

Layout

Results

Constraint identified: the “Compressor securing” station maintains an average utilization of approximately 95.06%, acting as the dominant bottleneck.

WIP behavior: for processing times greater than approximately 88 s, WIP increases steadily; 86.8 s acts as a transition point.

Key result in Layout 2: with an average processing time of 86.8 s at the constraint:

- Throughput: 316 units/day (39.50 units/hour)
- WIP: stabilizes around 16 units

Conclusions

Conclusion 1

The objective was achieved. The redesign made it possible to identify and validate a specialized layout with significant operational improvements, highlighting Layout 2.

Conclusion 2

The base system meets demand but relies on demanding operational conditions due to the constraint. Layout 2 improves flow stability and reduces WIP without compromising production.

Conclusion 3

The integration of Lean Manufacturing and simulation constitutes a robust approach for redesigning production systems.

Conclusion 4

Statistical validation confirms that differences between scenarios are significant ( $p < 0.05$ ).

Contact

Mateo Sebastián Echeverria Peña  
mateo.echeverria@ucuenca.edu.ec



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